Media, Technology, and the Race to Publish

Juan Cruz Llambias & Matthew J. H. Murphy
University of Pennsylvania

November 17, 2025

Introduction and Motivation

- Recent technological progress has changed the anatomy of news production:
    - The transition from print media to online news and social media distribution much easier.
    - Generative AI has decreased costs to the processing and creation of articles.
- Although these changes have made it cheaper to produce and distribute information, consumers increasingly believe that the quality of news has decreased.
- Previous work has studied misinformation and bias; we take a different approach.

Can technological changes affect the coverage decisions of news-producing firms?

News Production as a Race

We focus on a key insight: information production is a race. We build a model where:

- Firms compete to be the first to publish a “deep” story, with the outside option of publishing “simple” stories.
- Firms choose when to publish the “deep” story, either when it is fully completed or it is partially researched:
  - Publishing early hedges against competitors also publishing early.
  - But, waiting saves on “sharing” costs: writing, distributing, and advertising.

Overview of Results

We find that two major technological forces push firms' behaviour in opposite directions:

- Decreasing **research** costs makes firms more willing to cover deep stories, and to wait longer to publish stories until they are complete.
- Decreasing **publishing** costs shifts firms to cover simple stories, and to publish preliminary deep stories.

Overview of Results

We find that two major technological forces push firms' behaviour in opposite directions:

- Decreasing **research** costs makes firms more willing to cover deep stories, and to wait longer to publish stories until they are complete.
- Decreasing **publishing** costs shifts firms to cover simple stories, and to publish preliminary deep stories.

We also study how firms' information during the race impacts their decisions:

- When preliminary progress is hidden, firms are not more willing to cover deep stories, but they more frequently delay publication until stories are complete.

Related Literature

Empirical literature on news media:

- Franceschelli (2011), Cagé, Hervé, and Viaud (2020), Bhuller et al. (2024)

The effect of bias/misinformation on journalism quality:

- Bisceglia (2023), Gentzkow and Shapiro (2010), Che and Mierendorff (2019)

Methodologically most similar to the research race and patent literature:

- **Chatterjee, Das, and Dong (2025)**, Song and Zhao (2021), Kim and Tomaino (2024)
- Three major changes relative to these models: (i) endogenous outside option (simple story), (ii) publishing costs, (iii) non-linear production costs.

Model

Model: Firms

- Time is continuous $t \in [0, \infty)$.
- There are two firms: $i \in \{1, 2\}$ competing to publish news in continuous time.
- There are two types of news stories: a "deep" story d and a "simple" story s.

At each moment in time, the firms choose $a_i \in A_i = \mathbb{R}_+ \times [0, 1] \times \{0, 1\}$.

(i) How much effort to invest in researching $\lambda \in \mathbb{R}_+$;

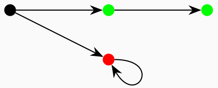
(ii) What fraction of their time is spent on the deep story $e \in [0, 1]$;

(iii) Whether to publish any information that they have: $p \in \{0, 1\}$.

Model: Researching and Publishing

When a firm invests λ in research during $[t, t + dt)$, it pays a flow cost $\alpha\, C_r(\lambda)dt$, and with probability λdt the firm achieves a "breakthrough".

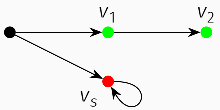
- Simple stories are repeatable, and require only one breakthrough.
- The deep story is unique, and requires two *sequential* breakthroughs.



Model: Researching and Publishing

When a firm invests λ in research during $[t, t + dt)$, it pays a flow cost $\alpha\, C_r(\lambda)dt$, and with probability λdt the firm achieves a "breakthrough".

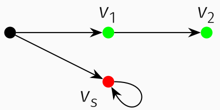
- Simple stories are repeatable, and require only one breakthrough.
  - Simple stories generate payoff v_s when published.
- The deep story is unique, and requires two *sequential* breakthroughs.
  - Publishing the first part yields v_1, the second part v_2.
  - Publishing the two together yields $v_1 + v_2$.
  - When part one is published, both firms can work on part two.



Model: Researching and Publishing

When a firm invests λ in research during $[t, t + dt)$, it pays a flow cost $\alpha\, C_r(\lambda)dt$, and with probability λdt the firm achieves a "breakthrough".

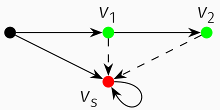
- Simple stories are repeatable, and require only one breakthrough.
  - Simple stories generate payoff v_s when published.
- The deep story is unique, and requires two *sequential* breakthroughs.
  - Publishing the first part yields v_1, the second part v_2.
  - Publishing the two together yields $v_1 + v_2$.
  - When part one is published, both firms can work on part two.
- Whenever the firm publishes a story, it pays a "publishing cost" $c_p > 0$.



Model: Researching and Publishing

When a firm invests λ in research during $[t, t + dt)$, it pays a flow cost $\alpha\ C_r(\lambda)dt$, and with probability λdt the firm achieves a "breakthrough".

- Simple stories are repeatable, and require only one breakthrough.
  - Simple stories generate payoff v_s when published.
- The deep story is unique, and requires two *sequential* breakthroughs.
  - Publishing the first part yields v_1, the second part v_2.
  - Publishing the two together yields $v_1 + v_2$.
  - When part one is published, both firms can work on part two.
- Whenever the firm publishes a story, it pays a "publishing cost" $c_p > 0$.



Model: Researching and Publishing

When a firm invests λ in research during $[t, t + dt)$, it pays a flow cost $\alpha\, C_r(\lambda)dt$, and with probability λdt the firm achieves a "breakthrough".

- Simple stories are repeatable, and require only one breakthrough.
  - Simple stories generate payoff v_s when published.
- The deep story is unique, and requires two *sequential* breakthroughs.
  - Publishing the first part yields v_1, the second part v_2.
  - Publishing the two together yields $v_1 + v_2$.
  - When part one is published, both firms can work on part two.
- Whenever the firm publishes a story, it pays a "publishing cost" $c_p > 0$.

Assumption

$C_r : \mathbb{R}_+ \to \mathbb{R}_+$ is strictly convex, with $C_r'(0) = 0$ and $C_r''' \geq 0$.

Model: Payoffs

Firms discount the future at rate r, and aim to maximize their expected discounted payoff from researching and publishing stories:

$$\int_0^\infty e^{-rt}\Big(\delta_{\{\text{Publish story of value } v_t \text{ at time } t\}} \cdot \big(v_t - c_p\big) - C_r(\lambda_t)\Big)dt$$

$$v_1 \in \{v_s, v_1, v_2, v_1 + v_2\}$$

Model: Firms' information

Firms always observe when the other publishes (part of) the deep story. We focus on two specifications:

- Firms observe each others' breakthroughs.
- Firms only observe publications.

These two assumptions lead to very different settings, but similar qualitative conclusions.

Model: Firms' information

Firms always observe when the other publishes (part of) the deep story. We focus on two specifications:

- Firms observe each others' breakthroughs.
- Firms only observe publications.

These two assumptions lead to very different settings, but similar qualitative conclusions.

When firms' breakthroughs are observable, we need an additional assumption on timing:

- Since breakthroughs are observable, we assume that if firm i achieves a breakthrough at time t, it has first right to publish.

Observable Breakthroughs

Solution Concept and Relevant States

While histories are generally complicated, we focus on Markov strategies.

- The simple story's publication does not affect the underlying state.

There are six states, reflecting progress along the deep story:

Solution Concept and Relevant States

While histories are generally complicated, we focus on Markov strategies.

- The simple story's publication does not affect the underlying state.

There are six states, reflecting progress along the deep story:

- p_2: both parts of the deep story have been published.
- p_1: the first breakthrough of the deep story has been published.

Solution Concept and Relevant States

While histories are generally complicated, we focus on Markov strategies.

- The simple story's publication does not affect the underlying state.

There are six states, reflecting progress along the deep story:

- p_2: both parts of the deep story have been published.
- p_1: the first breakthrough of the deep story has been published.
- $b_{0,0}$: neither firm has achieved a breakthrough for the deep story.
- $b_{1,1}$: both firms have achieved a breakthrough.

Solution Concept and Relevant States

While histories are generally complicated, we focus on Markov strategies.

- The simple story's publication does not affect the underlying state.

There are six states, reflecting progress along the deep story:

- p_2: both parts of the deep story have been published.
- p_1: the first breakthrough of the deep story has been published.
- $b_{0,0}$: neither firm has achieved a breakthrough for the deep story.
- $b_{1,1}$: both firms have achieved a breakthrough.
- $b_{1,0}$: the first firm has achieved a breakthrough, the second hasn't.
- $b_{0,1}$: the second firm has achieved a breakthrough, the first hasn't.

Determining Equilibrium Investment: $\omega = p_2$

Hold fixed the publishing and research decisions of each firm. Let $V_i(\omega)$ denote the value of firm i in state ω. Consider first the terminal state p_2:

$$V_i(p_2) = \max_{\lambda_i} \left\{ \lambda_i \left(v_s - c_p \right) dt + (1 - r dt) V_i(p_2) - \alpha C_r(\lambda) \cdot dt \right\}$$

Determining Equilibrium Investment: $\omega = p_2$

Hold fixed the publishing and research decisions of each firm. Let $V_i(\omega)$ denote the value of firm i in state ω. Consider first the terminal state p_2:

$$
\begin{aligned}
V_i(p_2) &= \max_{\lambda_i} \left\{ \lambda_i \left(v_s - c_p \right) dt + (1 - r dt) V_i(p_2) - \alpha C_r(\lambda) \cdot dt \right\} \\
&= \max_{\lambda} \frac{\lambda (v_s - c_p) - \alpha C_r(\lambda)}{r}
\end{aligned}
$$

Since the simple story is repeatable, the other firm's decisions do not affect firm i's value.

Determining Equilibrium Investment: $\omega = b_{1,1}$

In earlier states, firms' choices depend on the other's choices. Consider $\omega = b_{1,1}$, conditional on no publishing and on working solely on the deep story.

$$V_i(b_{1,1};\lambda_{-i}) = \max_{\lambda_i} \Big\{ \lambda_i \big(v_1 + v_2 - c_p + V_i(p_2) - V_i(b_{1,1};\lambda_{-i}) \big) dt \\ - \alpha C_r(\lambda_i) dt + (1 - r dt) V_i(b_{1,1};\lambda_{-i}) \Big\}$$

Determining Equilibrium Investment: $\omega = b_{1,1}$

In earlier states, firms' choices depend on the other's choices. Consider $\omega = b_{1,1}$, conditional on no publishing and on working solely on the deep story.

$$V_i(b_{1,1};\lambda_{-i}) = \max_{\lambda_i}\Big\{\lambda_i\big(v_1 + v_2 - c_p + V_i(p_2) - V_i(b_{1,1};\lambda_{-i})\big)\,dt + \lambda_{-i}\big(V_i(p_2) - V_i(b_{1,1};\lambda_{-i})\big)\,dt$$
$$- \alpha C_r(\lambda_i)dt + (1 - r dt)\,V_i(b_{1,1};\lambda_{-i})\Big\}$$

Determining Equilibrium Investment: $\omega = b_{1,1}$

In earlier states, firms' choices depend on the other's choices. Consider $\omega = b_{1,1}$, conditional on no publishing and on working solely on the deep story.

$$V_i(b_{1,1}; \lambda_{-i}) = \max_{\lambda_i} \frac{\lambda_i(v_1 + v_2 - c_p + V_i(p_2)) + \lambda_{-i} V_i(p_2) - \alpha C_r(\lambda_i)}{r + \lambda_i + \lambda_{-i}}$$

Determining Equilibrium Investment: $\omega = b_{1,1}$

In earlier states, firms' choices depend on the other's choices. Consider $\omega = b_{1,1}$, conditional on no publishing and on working solely on the deep story.

$$V_i(b_{1,1}; \lambda_{-i}) = \max_{\lambda_i} \frac{\lambda_i(v_1 + v_2 - c_p + V_i(p_2)) + \lambda_{-i} V_i(p_2) - \alpha C_r(\lambda_i)}{r + \lambda_i + \lambda_{-i}}$$

Get first-order condition

$$(r + \lambda_i + \lambda_{-i})\alpha C_r'(\lambda_i) - \alpha C_r(\lambda_i) = (r + \lambda_{-i})(v_1 + v_2 - c_p) + rV_i(p_2)$$

If firm i wants to work on the deep story, then $v_1 + v_2 - c_p > \alpha C'(\lambda_i) > 0$, so as λ_{-i} increases, firm i's optimal investment also increases.

- This implies that all equilibria investments in $b_{1,1}$ are symmetric.

Pinning-down Investment

$C_r''' \geq 0$ implies there are at most two solutions to the equation. Moreover, stability eliminates all but one possible equilibrium.

Lemma

Fixing firm publishing and coverage behaviour, there are at most two investment equilibria at each state.

- In all symmetric states ($\omega \neq b_{0,1}, b_{1,0}$), all equilibria are symmetric ($\lambda_1 = \lambda_2$).
- In each state, only one investment equilibrium is Evolutionarily Stable.

Asymmetric States

Backing-out Publishing and Research Decisions

With investment behaviour pinned-down, the firm can now decide when to publish and when give up on the deep story. Consider the choice at $b_{1,0}$:

- If the firm publishes immediately, its value will be $v_1 - c_p + V(p_1)$.
- If it chooses not to publish, its value becomes $V(b_{1,0})$.

I.e., it will publish immediately if and only if $V(b_{1,0}) < v_1 - c_p + V(p_1)$.

Backing-out Publishing and Research Decisions

With investment behaviour pinned-down, the firm can now decide when to publish and when give up on the deep story. Consider the choice at $b_{1,0}$:

- If the firm publishes immediately, its value will be $v_1 - c_p + V(p_1)$.
- If it chooses not to publish, its value becomes $V(b_{1,0})$.

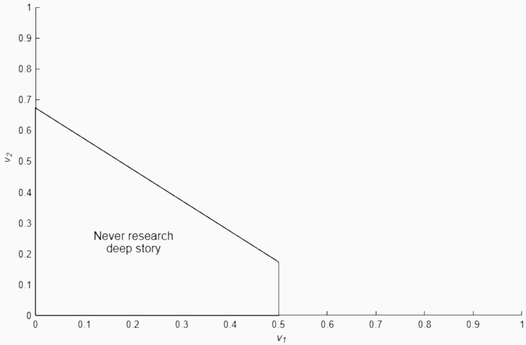
I.e., it will publish immediately if and only if $V(b_{1,0}) < v_1 - c_p + V(p_1)$.

In state $b_{0,0}$, firm i works on the deep story if and only if

$$v_s - c_p + V(b_{0,0}) < \max\left\{V(b_{1,0}), v_1 - c_p + V(p_1)\right\}.$$

Theorem

There is a unique Evolutionarily Stable Markov Perfect Equilibrium. Fixing v_s, c_p, α and r, the behaviour or firms is decided by v_1 and v_2. There are five possibilities for firm behaviour.

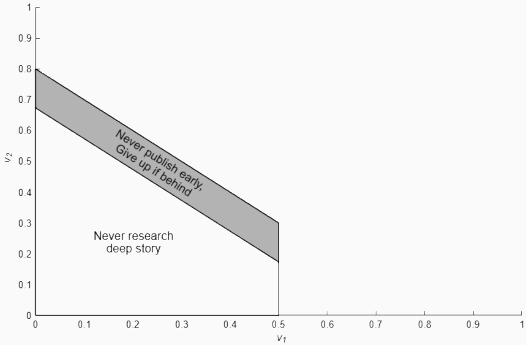


$C_r(\lambda) = \frac{1}{2}\lambda^2$, $v_s = 0.5$, $c_p = 0.3$, $\alpha = 1$, $r = 0.1$.

Equilibrium Characterization

Theorem

There is a unique Evolutionarily Stable Markov Perfect Equilibrium. Fixing v_s, c_p, α and r, the behaviour or firms is decided by v_1 and v_2. There are five possibilities for firm behaviour.

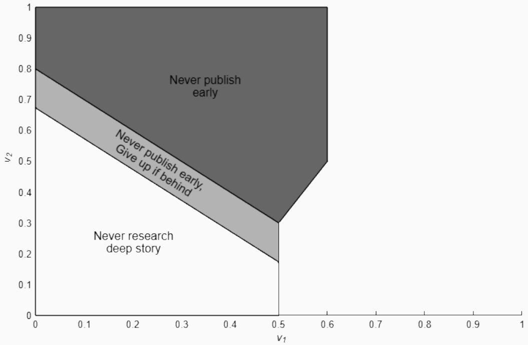


$C_r(\lambda) = \frac{1}{2}\lambda^2$, $v_s = 0.5$, $c_p = 0.3$, $\alpha = 1$, $r = 0.1$.

Equilibrium Characterization

Theorem

There is a unique Evolutionarily Stable Markov Perfect Equilibrium. Fixing v_s, c_p, α and r, the behaviour or firms is decided by v_1 and v_2. There are five possibilities for firm behaviour.

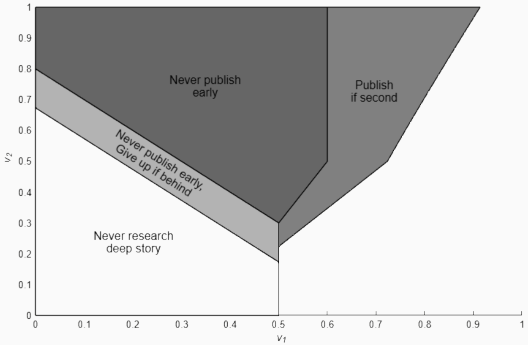


$C_r(\lambda) = \frac{1}{2}\lambda^2$, $v_s = 0.5$, $c_p = 0.3$, $\alpha = 1$, $r = 0.1$.

Equilibrium Characterization

Theorem

There is a unique Evolutionarily Stable Markov Perfect Equilibrium. Fixing v_s, c_p, α and r, the behaviour or firms is decided by v_1 and v_2. There are five possibilities for firm behaviour.

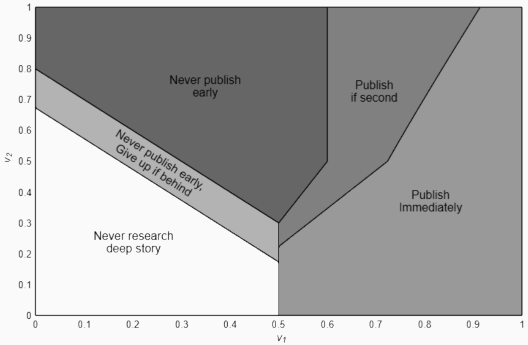


$C_r(\lambda) = \frac{1}{2}\lambda^2$, $v_s = 0.5$, $c_p = 0.3$, $\alpha = 1$, $r = 0.1$.

Equilibrium Characterization

Theorem

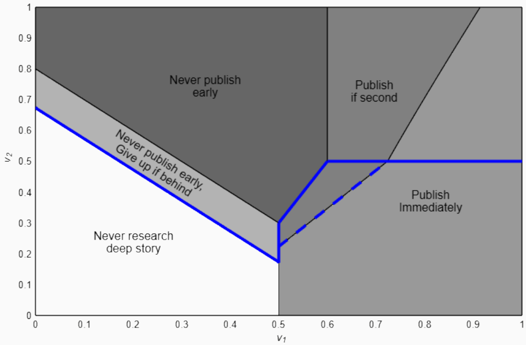
There is a unique Evolutionarily Stable Markov Perfect Equilibrium. Fixing v_s, c_p, α and r, the behaviour or firms is decided by v_1 and v_2. There are five possibilities for firm behaviour.



$C_r(\lambda) = \frac{1}{2}\lambda^2$, $v_s = 0.5$, $c_p = 0.3$, $\alpha = 1$, $r = 0.1$.

Theorem

There is a unique Evolutionarily Stable Markov Perfect Equilibrium. Fixing v_s, c_p, α and r, the behaviour or firms is decided by v_1 and v_2. There are five possibilities for firm behaviour.

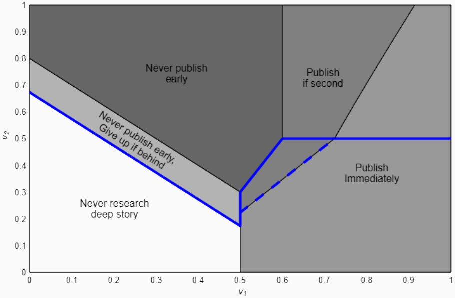


$C_r(\lambda) = \frac{1}{2}\lambda^2$, $v_s = 0.5$, $c_p = 0.3$, $\alpha = 1$, $r = 0.1$.

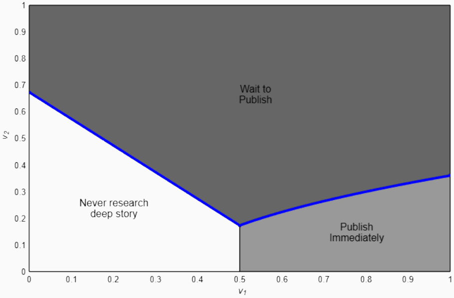
Comparison with the Monopolist

Consider what happens if one firm drops out.

- The monopolist may still choose to forgo the deep story, and may still publish early.



Competitive Equilibrium



Monopolist

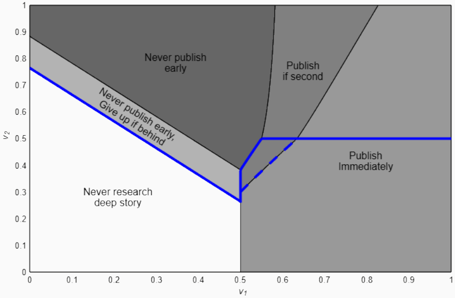
Comparative Statics

The effect of a decrease in α

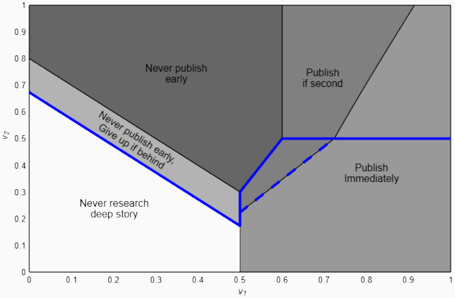
A decrease in research costs increases the benefit of waiting for the deep story.

Proposition

As α decreases, firms are more likely to cover the deep story, and are less likely to publish the deep story early.



$\alpha = 3$



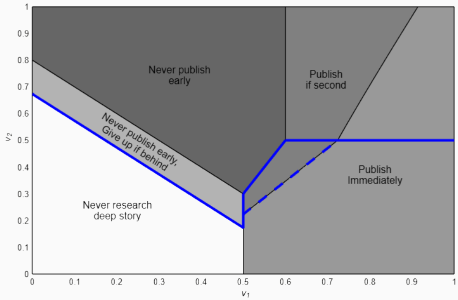
$\alpha = 1$

The effect of a decrease in c_p

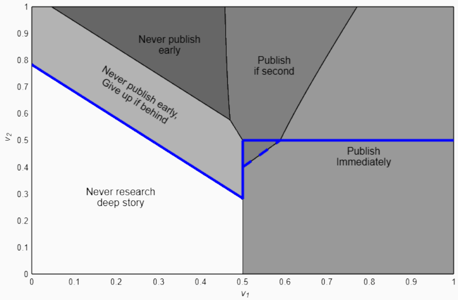
A decrease in publishing costs c_p, decreases the benefit of waiting to publish.

Proposition

As c_p decreases, firms are less likely to cover the deep story, and are more likely to publish the deep story early.



$c_p = 0.4$



$c_p = 0.3$

Unobserved Breakthroughs (Preliminary)

Modelling Differences

Firms' progress on the deep story is now hidden. We also assume that firms do not observe the other firm's level of investment λ or their choice of coverage e.

- Firms' now have belief $\mu \in [0, 1]$ that the other firm has achieved a breakthrough on the deep story.

We look for symmetric Markov Perfect Equilibria, where actions depend only on the (public) beliefs about the other firms' breakthroughs.

Assumption

Firm i does not see when firm j publishes a simple story.

Belief Dynamics

- $\lambda_0(\mu)$: investment of firm with no breakthrough, and belief μ. Similarly $\lambda_1(\mu)$.
- $P(\mu) \in [0, \infty]$: publication rate, conditional on having a breakthrough.

$$\frac{\partial}{\partial t}\mu = (1-\mu)(\underbrace{\lambda_0(\mu)}_{\text{inflow}} - \mu(\underbrace{\lambda_1(\mu) + P(\mu)}_{\text{outflow}}))$$

If $\frac{\partial}{\partial t}\mu = 0$, then we have reached a steady-state in beliefs. These will always be absorbing.

Value Functions and ODE

Let $V_0(\mu)$ denote the value of a firm with no breakthrough, with belief μ in the other firms' breakthrough (similarly $V_1(\mu)$. Functions λ_0, λ_1, P induce the following ODE:

$0 =$

When $\frac{\partial}{\partial t}\mu = 0$, then this problem becomes stationary.

Value Functions and ODE

Let $V_0(\mu)$ denote the value of a firm with no breakthrough, with belief μ in the other firms' breakthrough (similarly $V_1(\mu)$. Functions λ_0, λ_1, P induce the following ODE:

$$0 = \underbrace{\lambda_{0,i}(\mu)\,(V_1(\mu) - V_0(\mu))}_{\text{Own impact}} - C_r(\lambda_{0,i}(\mu)) - rV_0(\mu)$$

When $\frac{\partial}{\partial t}\mu = 0$, then this problem becomes stationary.

Value Functions and ODE

Let $V_0(\mu)$ denote the value of a firm with no breakthrough, with belief μ in the other firms' breakthrough (similarly $V_1(\mu)$. Functions λ_0, λ_1, P induce the following ODE:

$$0 = \underbrace{\lambda_{0,i}(\mu)\,(V_1(\mu) - V_0(\mu))}_{\text{Own impact}} - C_r(\lambda_{0,i}(\mu)) - rV_0(\mu) + \underbrace{\mu\Big(\lambda_{1,-i}(\mu)(V(p_2) - V_0(\mu)) + P_{-i}(\mu)(V(p_1) - V_0(\mu))\Big)}_{\text{Other's impact}}$$

When $\frac{\partial}{\partial t}\mu = 0$, then this problem becomes stationary.

Value Functions and ODE

Let $V_0(\mu)$ denote the value of a firm with no breakthrough, with belief μ in the other firms' breakthrough (similarly $V_1(\mu)$. Functions λ_0, λ_1, P induce the following ODE:

$$0 = \underbrace{\lambda_{0,i}(\mu)\,(V_1(\mu) - V_0(\mu))}_{\text{Own impact}} - C_r(\lambda_{0,i}(\mu)) - rV_0(\mu) + \underbrace{\mu\Big(\lambda_{1,-i}(\mu)(V(p_2) - V_0(\mu)) + P_{-i}(\mu)(V(p_1) - V_0(\mu))\Big)}_{\text{Other's impact}}$$

$$+ \quad \underbrace{\frac{\partial}{\partial\mu}V_0(\mu)}_{\Delta\text{ from }\Delta\text{ in }\mu} \quad \underbrace{\Big[(1-\mu)\Big(\lambda_{0,-i}(\mu) - \mu(\lambda_{1,-i}(\mu) + P_{-i}(\mu))\Big)\Big]}_{\text{Rate at which }\mu\text{ changes}}$$

When $\frac{\partial}{\partial t}\mu = 0$, then this problem becomes stationary.

Calculating Belief Dynamics in Equilibrium

Note: $V_1(\mu)$ is decreasing in μ. Consider the belief dynamics in equilibrium:

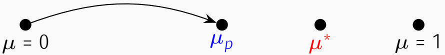
Calculating Belief Dynamics in Equilibrium

Note: $V_1(\mu)$ is decreasing in μ. Consider the belief dynamics in equilibrium:

- Ignoring switching and publishing, beliefs increase until μ^* with $\mu^* = \frac{\lambda_0(\mu^*)}{\lambda_1(\mu^*)}$.

Calculating Belief Dynamics in Equilibrium

Note: $V_1(\mu)$ is decreasing in μ. Consider the belief dynamics in equilibrium:



- Ignoring switching and publishing, beliefs increase until μ^* with $\mu^* = \frac{\lambda_0(\mu^*)}{\lambda_1(\mu^*)}$.
- If $V_1(\mu^*) < v_1 - c_p + V(p_1)$, this firm would rather publish $\Rightarrow$ not an equilibrium.
- For μ_p with $V_1(\mu_p) = v_1 - c_p + V(p_1)$ will have stochastic publishing.
  - The rate of publishing $P(\mu_p)$ ensures beliefs stay constant.

Calculating Belief Dynamics in Equilibrium

Note: $V_1(\mu)$ is decreasing in μ. Consider the belief dynamics in equilibrium:

- Ignoring switching and publishing, beliefs increase until μ^* with $\mu^* = \frac{\lambda_0(\mu^*)}{\lambda_1(\mu^*)}$.
- If $V_1(\mu^*) < v_1 - c_p + V(p_1)$, this firm would rather publish $\Rightarrow$ not an equilibrium.
- For μ_p with $V_1(\mu_p) = v_1 - c_p + V(p_1)$ will have stochastic publishing.
  - The rate of publishing $P(\mu_p)$ ensures beliefs stay constant.
- Can similarly deal with the issue of wanting to switch to the simple story: The binding constraint is $V_1(\mu_g) = v_s - c_p + V(p_2)$.
  - Will split time between the simple and deep stories to keep beliefs constant.

Equilibrium

Also need to make sure that beliefs move from zero: firms must be willing to work on the deep story when $\mu = 0$, and must not publish immediately.

Theorem

Any equilibrium will feature the same belief dynamic behaviour. Either

(i) Firms never work on the deep story $\mu \equiv 0$;

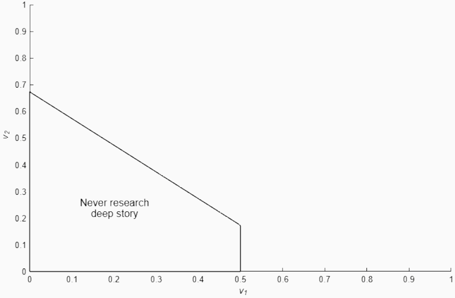
(ii) Firms always publish immediately $\mu \equiv 0$; or

(iii) Beliefs increase to some steady state $\mu^* \in (0, 1)$.

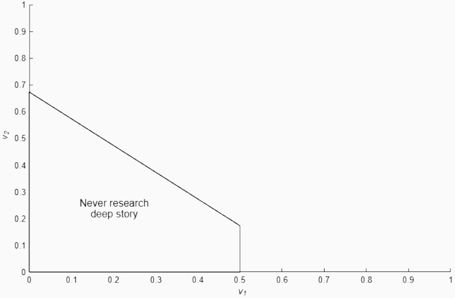
In total, there are five possibilities for firm behaviour.

Still in the process of ensuring that the equilibrium is unique for general cost functions.

Comparison with observable breakthroughs

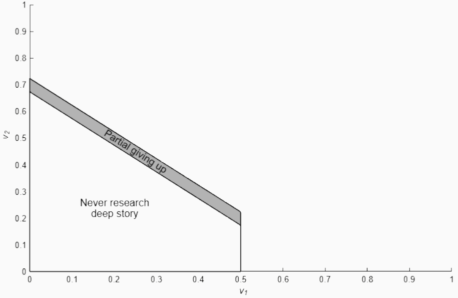


Unobservable Breakthroughs

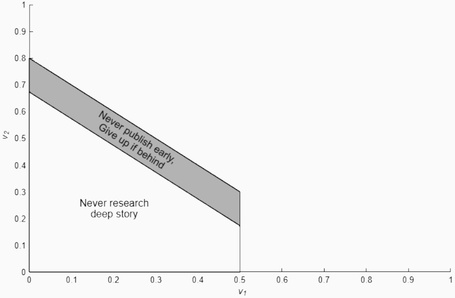


Observable Breakthroughs

Comparison with observable breakthroughs

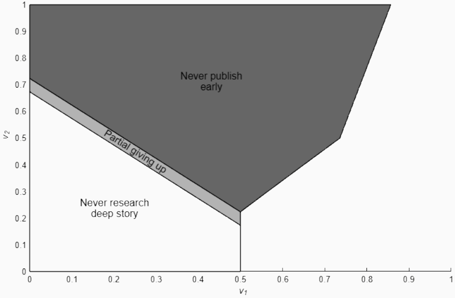


Unobservable Breakthroughs

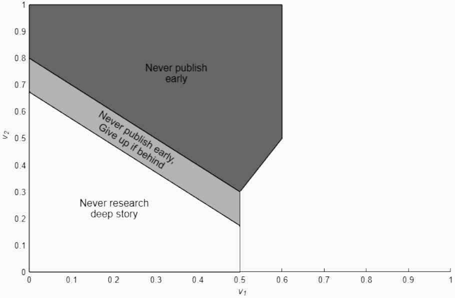


Observable Breakthroughs

Comparison with observable breakthroughs

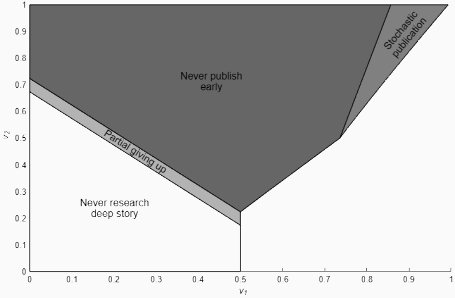


Unobservable Breakthroughs

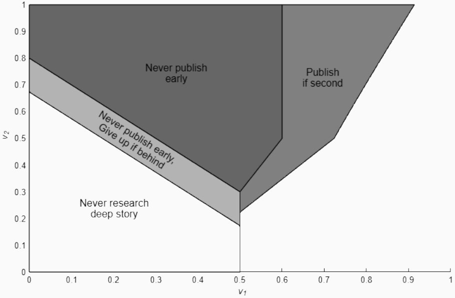


Observable Breakthroughs

Comparison with observable breakthroughs

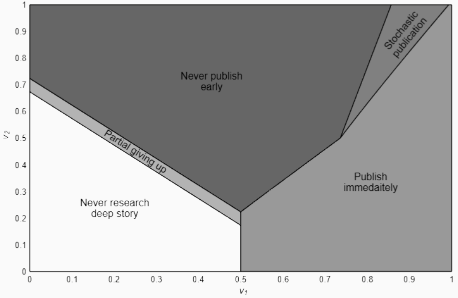


Unobservable Breakthroughs

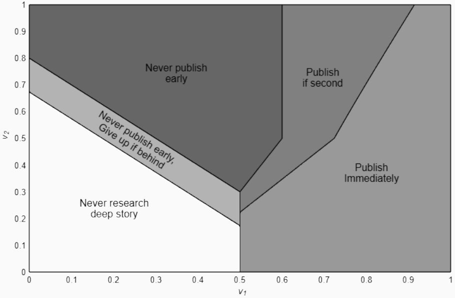


Observable Breakthroughs

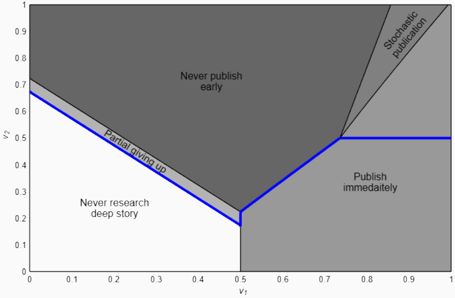
Comparison with observable breakthroughs



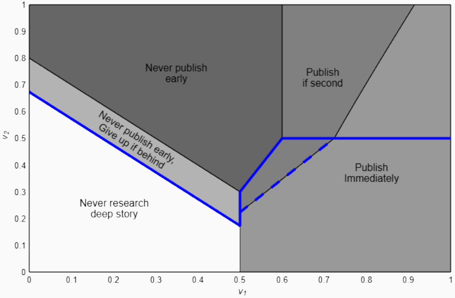
Unobservable Breakthroughs



Observable Breakthroughs

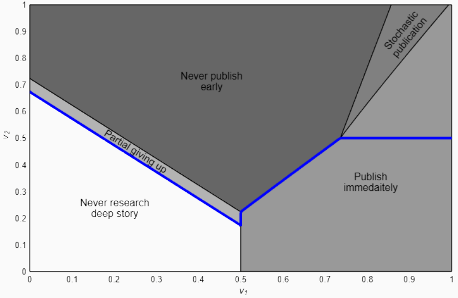


Unobservable Breakthroughs

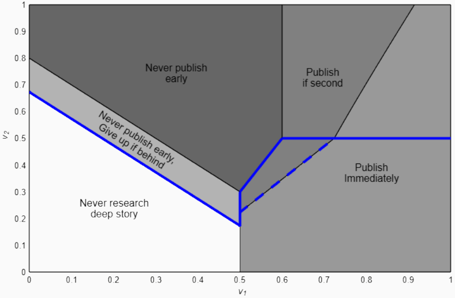


Observable Breakthroughs

Comparison with observable breakthroughs



Unobservable Breakthroughs



Observable Breakthroughs

Proposition

Compared with when breakthroughs are observable, firms publish immediately upon a breakthrough less frequently, and are more willing to always wait until the deep story is completed to publish.

Future Directions

- Incorporate an appropriate metric of “quantity” of information.
    - Whether investment increases or decreases depends on the region *and* the state.
    - No clear effect on “aggregate” investment.
- Study the effect of copying on firms’ behaviour
    - If copiers have to pay a similar publishing cost, then this will amplify the existing forces.
- We would appreciate insight on how to frame the model, if there are other applications where you see its insights being relevant, other directions we could take the project, etc.

Conclusion

We develop a model where firms race to produce and distribute information. We find

- Decreasing research costs encourages firms to research deep stories; but
- Decreasing publishing costs has the opposite effects.
- Firms' ability to hide breakthroughs also encourages them to delay publication by improving information rents.

Asymmetric State $b_{1,0}$: Publishing

Here, the firms' investments will *not* be the same. First, suppose that the lagging firm publishes if it catches up. We have value functions

$$V_1(1,0) = \frac{\lambda_1(v_1 + v_2 - c_p + V_1(p_2)) + \lambda_2 V_1(p_1) - \alpha C_r(\lambda_1)}{r + \lambda_1 + \lambda_2}$$

$$V_2(1,0) = \frac{\lambda_1 V_2(p_2) + \lambda_2(v_1 - c_p + V_2(p_1)) - \alpha C_r(\lambda_2)}{r + \lambda_1 + \lambda_2}$$

and first-order conditions:

$$(r + \lambda_1 + \lambda_2)C_r'(\lambda_1) - C_r(\lambda_1) = (r + \lambda_2)(v_1 + v_2 - c_p + V(p_2)) - \lambda_2 V(p_1)$$

$$(r + \lambda_1 + \lambda_2)C_r'(\lambda_2) - C_r(\lambda_2) = (r + \lambda_1)(v_1 - c_p + V(p_1)) - \lambda_1 V(p_2).$$

Back

Asymmetric State $b_{1,0}$: No Publishing

Here, the firms' investments will *not* be the same. Now, suppose that the lagging firm does not publish if it catches up.We get value functions

$$V_1(1,0) = \frac{\lambda_1(v_1 + v_2 - c_p + V_1(p_2)) + \lambda_2 V_1(b_{1,1}) - \alpha C_r(\lambda_1)}{r + \lambda_1 + \lambda_2}$$

$$V_2(1,0) = \frac{\lambda_1 V_2(p_2) + \lambda_2 V_2(b_{1,1}) - \alpha C_r(\lambda_2)}{r + \lambda_1 + \lambda_2}$$

and first-order conditions:

$$(r + \lambda_1 + \lambda_2)C_r'(\lambda_1) - C_r(\lambda_1) = (r + \lambda_2)(v_1 + v_2 - c + V(p_2)) - \lambda_2 V(b_{1,1})$$

$$(r + \lambda_1 + \lambda_2)C_r'(\lambda_2) - C_r(\lambda_2) = (r + \lambda_1)V(b_{1,1}) - \lambda_1 V(p_2).$$

Back